A WEIHRAUCH SEPARATION BETWEEN 1-GENERICITY AND THE FINITE INTERSECTION PRINCIPLE

PATRIZIO CINTIOLI

Abstract. This paper shows that the problem of producing a Cohen 1-generic real relative to a given oracle is not Weihrauch reducible to the finite intersection principle (FIP). More strongly, no nontrivial computable FIP instance admits a single Turing functional that produces an absolute 1-generic real from every enumeration of every maximal centered set of indices. The argument combines a 1-generic real that computes no perfect tree all of whose paths are 1-generic with local constancy and a parametrization of maximal centered sets. This answers Question 9.10.25 of Dzhafarov and Mummert.

1. Introduction

The *finite intersection principle* (FIP) asks, for every nontrivial countable family of subsets of $\mathbb{N}$, for an inclusion-maximal subfamily in which every finite collection of at least two distinct members has nonempty intersection. Its relationship with Cohen genericity has been studied in both computability theory and reverse mathematics. Cholak, Downey, and Igusa [2] proved that a Turing degree is FIP-bounding if and only if it computes a Cohen 1-generic; they also established the equivalence of FIP and 1-GEN over RCA_0. Immediately after Theorem 1.6, they contrasted the uniform extraction available in the earlier Δ^0_2 setting with their unrestricted argument, which uses infinitely many candidate Turing functionals. They also reported preliminary evidence that no single uniform extraction procedure could work in the general case. The present result makes this nonuniformity obstruction precise in the setting of Weihrauch reducibility.

The uniform computational content of 1-genericity was investigated by Brattka, Hendtlass, and Kreuzer [1]. The Weihrauch-theoretic study of FIP also appears in the work of Dzhafarov, Solomon, and Yokoyama [6], concerning the first-order parts of computational problems. The question here is whether an entire generic real can be extracted uniformly. Dzhafarov and Mummert [5, Question 9.10.25] explicitly asked whether 1-genericity is Weihrauch reducible to FIP. The present paper gives a negative answer. The obstruction has a stronger form: a single postprocessor cannot succeed on every solution presentation of *any fixed* nontrivial computable FIP instance.

Let GEN be the class of absolute Cohen 1-generic reals in 2^ω. Recall that $G \in 2^\omega$ is 1-generic if, for every c.e. set $U \subseteq 2^{<\omega}$, at least one of the following alternatives holds:

$$(\exists n)\, G \upharpoonright n \in U, \qquad (\exists n)(\forall \tau \succeq G \upharpoonright n)\, \tau \notin U.$$

A nontrivial computable family is a sequence $\mathcal{A} = (A_i)_{i\in\mathbb{N}}$ of subsets of $\mathbb{N}$, with the relation $x \in A_i$ uniformly computable in i, x, and with at least one nonempty A_i.

Theorem 1. *For no nontrivial computable family $\mathcal{A}$ does there exist a Turing functional Φ such that, for every maximal centered set of indices M and every enumeration J of all of M, the real Φ^J is total and belongs to GEN.*

An enumeration is a total sequence, possibly with repetitions, whose range is exactly the set being enumerated. The required functional must work for all orders and repetitions: no permutation-invariance of its output is assumed.

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As an immediate consequence of Theorem 1, proved in Section 7, we obtain

$$1\text{-}\mathsf{GEN} \not\leq_{\mathrm{W}} \mathsf{FIP},$$

where 1-GEN is the multivalued problem that assigns to an oracle X the collection of reals that are 1-generic relative to X. This distinction between reverse-mathematical implication and Weihrauch reducibility is essential: the latter requires a single uniform pair of preprocessing and postprocessing procedures.

The proof of the stronger theorem makes essential use of the following result.

Theorem 2 (Hirschfeldt–Jockusch–Schupp, Theorem 7.23)**.** *There exists a* 1*-generic real* G *that computes no perfect tree all of whose paths are* 1*-generic.*

This is [7, Theorem 7.23]; its proof uses Kumabe's construction of a 1-generic degree with a strong minimal cover [8]. The perfect trees below are represented by splitting maps as in [7, Definition 3.20], with computability relative to the designated oracle.

The main idea is to parametrize centered sets by reals so that 1-generic parameters yield maximal centered sets and any finite centered extension can be realized above any parameter prefix. Theorem 2 then provides a parameter for which our hypothetical postprocessor is locally constant on one cone of enumerations. Genericity promotes this local constancy to a neighborhood of the parameter, and the extension property transfers it to every maximal centered set extending a fixed finite condition. A computable-output argument finally yields a computable 1-generic, a contradiction.

2. Finite conditions and representation of solutions

The relevant finite conditions are finite sets of indices whose corresponding family members have nonempty intersection. A solution is represented by an enumeration of a maximal collection of such compatible indices; the order and repetitions are unrestricted. The formulation in [5, Definition 9.10.17] requires nonempty intersections for all nonempty finite sets of indices, including singletons. The centered sets used here satisfy this stronger condition. Consequently, the solutions produced by the construction are valid under either convention.

Fix $\mathcal{A}$, and set

$$\mathcal{C} = \left\{ F \subseteq_{\mathrm{fin}} \mathbb{N} : \bigcap_{i \in F} A_i \neq \varnothing \right\},$$

with the convention that $\varnothing \in \mathcal{C}$. The class $\mathcal{C}$ is c.e. and downward closed. The requirement on singletons makes every selected family member nonempty; this stronger finite condition is convenient for the proof and still produces solutions to the usual FIP problem. For $F \neq \varnothing$, a positive witness to $F \in \mathcal{C}$ is a number w belonging to every A_i with $i \in F$.

A set $M \subseteq \mathbb{N}$ is *centered* if each of its finite subsets belongs to $\mathcal{C}$. It is *maximal* if it is inclusion-maximal among centered sets.

Every centered set extends to a maximal one. It suffices to consider all indices in increasing order and add each index whenever doing so preserves centeredness. This is an existence construction, not a computable algorithm.

For a maximal set M, define

$$E_M = \{J \in \omega^\omega : \mathrm{ran}(J) = M\}.$$

A maximal centered set is nonempty because the family is nontrivial. Every finite string with values in M can be extended to an element of E_M, even if M is finite, since repetitions are allowed.

For a finite string σ, let $[\sigma]$ denote the cylinder of sequences extending it, in Baire or Cantor space according to the context. When we write that Φ^σ determines a binary string u, we mean a finite computation of the first $|u|$ bits that uses only positions available in σ.

We write

$$\mathrm{Split}_\Phi(\tau_0, \tau_1)$$

for the property that there exist incompatible binary strings u_0, u_1 such that Φ^{τ_i} determines u_i, for $i = 0, 1$. This property is c.e.

3. Local constancy and computable output

The first lemma converts constancy along the enumerations of each maximal extension of one finite condition into a computable output. The construction does not require a computable maximal centered set.

Lemma 3. *Suppose that Φ is total, with values in 2^ω, on all enumerations of all maximal centered sets of $\mathcal{A}$. Let $c \in \mathcal{C}$ be nonempty, and let $\sigma \in \omega^{<\omega}$ satisfy $\mathrm{ran}(\sigma) \subseteq c$. If, for every maximal $M \supseteq c$, the restriction of Φ to $E_M \cap [\sigma]$ is constant, then some output of Φ on a maximal solution is computable. The constant value may depend on M.*

Proof. Let q_0 be the concatenation of σ with the increasing list of the elements of c; thus $\mathrm{ran}(q_0) = c$.

Computably construct increasing strings q_n, with centered ranges, such that $|q_n| \geq n$ and Φ^{q_n} determines at least n bits.

Given q_n, search for $q_{n+1} \succeq q_n$ with centered range, length at least $n+1$, and a convergent computation of at least $n+1$ bits. The search is c.e., using a positive witness to $\mathrm{ran}(q_{n+1}) \in \mathcal{C}$.

The search terminates: $\mathrm{ran}(q_n)$ extends to a maximal set N, and an enumeration of N beginning with q_n makes Φ total. A sufficiently long initial segment of that enumeration supplies the required witness.

The computations are compatible because the input strings extend one another. The limit of their outputs is a computable real g.

The set

$$S = \bigcup_n \mathrm{ran}(q_n)$$

is centered. Fix a maximal $M \supseteq S$. For every n, there exists $J_n \in E_M$ extending q_n. All these J_n extend σ, so their outputs Φ^{J_n} coincide by hypothesis. The common value extends $g \upharpoonright n$ for every n, and hence equals g. □

Remark. We do not construct a computable maximal solution. We construct a c.e. centered set, extend it abstractly to a maximal set, and use constancy on the cone of enumerations of that maximal set. The finite constants c, σ can be hard-coded into the program computing g.

4. From output variation to a perfect tree

If the outputs vary above every finite enumeration stem, this variation can be organized into a perfect binary tree of outputs. Positive access to the given enumeration suffices to compute the tree.

Lemma 4. *Let $M \neq \varnothing$, and suppose that Φ is total on E_M, with values in a class $\mathcal{Q} \subseteq 2^\omega$. At least one of the following alternatives holds:*

1. *there exists a string σ with values in M such that Φ is constant on $E_M \cap [\sigma]$;*
2. *every $J \in E_M$ computes a perfect tree $T \subseteq 2^{<\omega}$ with $[T] \subseteq \mathcal{Q}$.*

Proof. Suppose that the first alternative fails, and fix $J \in E_M$. Above every finite string p with values in M, there are two enumerations of M beginning with p and having distinct outputs. Hence there exist two finite extensions p_0, p_1 of p, still with values in M, such that $\mathrm{Split}_\Phi(p_0, p_1)$.

This pair can be searched for effectively using J: membership of the finitely many values of p_0, p_1 in M can be verified positively using J, while the splitting is witnessed by finite computations. The characteristic function of M is not used.

Using J as an oracle, construct input strings p_α and output strings u_α, for $\alpha \in 2^{<\omega}$, with the following properties:

(1) if $\alpha \preceq \beta$, then $p_\alpha \preceq p_\beta$ and $u_\alpha \preceq u_\beta$;

(2) $u_{\alpha^\frown 0}$ and $u_{\alpha^\frown 1}$ are incompatible and properly extend u_α;
(3) Φ^{p_α} determines u_α, and $|u_\alpha| \geq |\alpha|$;
(4) in passing from level n to level $n+1$, before searching for a splitting, append $J(n)$ to every input stem.

Start with $p_\varnothing = u_\varnothing = \varnothing$. The search described above guarantees each step. The children's outputs can be extended as far as needed using totality on completions in E_M.

For every $B \in 2^\omega$, the limit

$$f_B = \bigcup_n p_{B\restriction n}$$

is an enumeration of all of M: it uses only elements of M, includes $J(n)$ for every n, and has infinite length. Therefore

$$\bigcup_n u_{B\restriction n} = \Phi^{f_B} \in \mathcal{Q}.$$

The prefix closure of the strings u_α is a perfect tree T. It is computable in J, not merely c.e. in J: to decide whether a string v of length m belongs to T, it suffices to construct level m of the outputs and check whether v is a prefix of one of them. All these outputs have length at least m.

To see that every path through T is one of the outputs described above, put

$$L_n = \max\{|u_\alpha| : |\alpha| = n\}.$$

Every path through T extends exactly one of the strings u_α at level n. These choices are coherent across levels and therefore determine a unique branch $B \in 2^\omega$. The path is consequently equal to $\bigcup_n u_{B\restriction n} = \Phi^{f_B}$. Thus $[T] \subseteq \mathcal{Q}$.

Moreover, the map $S(\alpha) = u_\alpha$ directly satisfies the representation of perfect trees used in [7, Definition 3.20]: the children are incompatible and properly extend the parent. Thus no noncomputable conversion between representations is needed. □

5. A map from generics to maximal sets

Forcing with strings having centered ranges, and its connection to genericity, appear in [4, Section 2, Lemma 2.1]. Here we make explicit a total parametrization on 2^ω with the required local universality.

Lemma 5. *There exist a total functional $R \mapsto J_R$, from 2^ω to ω^ω, and a computable function $\xi \mapsto c_\xi \in \mathcal{C}$, defined on finite binary strings, with the following properties:*

(1) *$c_\xi \neq \varnothing$, and $c_\xi \subseteq c_\eta$ whenever $\xi \preceq \eta$;*
(2) *$M_R := \operatorname{ran}(J_R) = \bigcup_n c_{R\restriction n}$ is centered;*
(3) *if R is 1-generic, M_R is maximal;*
(4) *for every ξ and every $D \in \mathcal{C}$ with $D \supseteq c_\xi$, there exists $\eta \succeq \xi$ with $c_\eta = D$. In particular, $D \subseteq M_R$ for every $R \succeq \eta$.*

Proof. Find i_0 with $A_{i_0} \neq \varnothing$ by searching for a pair (i_0, w) with $w \in A_{i_0}$. Set $F_0 = \{i_0\}$.

Use a computable bijective coding of triples (D, w, r), where D is a finite set and r is a free parameter. If k codes (D, w, r), define

$$e(F,k) = \begin{cases} D, & F \subseteq D \text{ and } (\forall i \in D)\, w \in A_i, \\ F, & \text{otherwise.} \end{cases}$$

The function e is total and computable. If F is centered, $e(F,k)$ is a centered extension of F. Every $D \in \mathcal{C}$ with $F \subseteq D$ occurs as $e(F,k)$ for arbitrarily large values of k, by choosing a witness w and varying r.

Read the completed blocks of R in the form

$$0^{k_0}1\,0^{k_1}1\,0^{k_2}1\cdots$$

and update $F_{n+1} = e(F_n, k_n)$. For a finite string ξ, let c_ξ be the last F_n obtained from its completed blocks. Ignore any incomplete final block; in particular $c_\varnothing = F_0$.

To make J_R total for every R, concatenate, for $n = 0, 1, \ldots$, the increasing, finite, nonempty lists of elements of $c_{R\restriction n}$. This enumerates their union computably in R, even if R contains only finitely many 1's. The first two properties follow from monotonicity.

We verify local universality. If ξ ends with an incomplete block of a zeros, and $D \supseteq c_\xi$ is centered, choose $k \geq a$ with $e(c_\xi, k) = D$. Then

$$\eta = \xi^\frown 0^{k-a}1$$

satisfies $c_\eta = D$. The free parameter ensures that every prefix, even one ending in the middle of a block, allows every future centered extension.

For maximality, fix i and set

$$U_i = \{\xi : i \in c_\xi\}.$$

The set U_i is computable and upward closed under extension. If R is 1-generic, either it meets U_i, in which case $i \in M_R$, or there is a $\xi \prec R$ no extension of which belongs to U_i.

In the second case $c_\xi \cup \{i\} \notin \mathcal{C}$: otherwise, local universality would yield an extension of ξ in U_i. Thus i cannot be added to M_R. This proves maximality. □

6. Proof of the main theorem

Proof of Theorem 1. Suppose, toward a contradiction, that a nontrivial computable family $\mathcal{A}$ and a functional Φ satisfy the property ruled out by the statement.

Choose G as in Theorem 2. Apply Lemma 5, and put

$$J = J_G, \qquad M = M_G.$$

Then $J \leq_T G$, and M is maximal.

Apply Lemma 4 to M, Φ and $\mathcal{Q} = \mathsf{GEN}$. The second alternative is impossible: it would give a perfect tree of 1-generic reals computable in J, and hence in G. Thus there exists a finite string σ, with $\operatorname{ran}(\sigma) \subseteq M$, such that Φ is constant on $E_M \cap [\sigma]$.

For this σ, define

$$F_\sigma = \left\{ R \in 2^\omega : \begin{array}{l} \text{there do not exist } \tau_0, \tau_1 \succeq \sigma \text{ such that} \\ \operatorname{ran}(\tau_0) \cup \operatorname{ran}(\tau_1) \subseteq M_R \text{ and } \operatorname{Split}_\Phi(\tau_0, \tau_1) \end{array} \right\}.$$

To make its complexity explicit, put

$$V_\sigma = \left\{ \zeta \in 2^{<\omega} : \begin{array}{l} \text{there exist } \tau_0, \tau_1 \succeq \sigma \text{ such that} \\ \operatorname{ran}(\tau_0) \cup \operatorname{ran}(\tau_1) \subseteq c_\zeta \text{ and } \operatorname{Split}_\Phi(\tau_0, \tau_1) \end{array} \right\}.$$

The set V_σ is c.e. and upward closed under extension: c_ζ is finite and computable, and a splitting has finite computational witnesses. By monotonicity of the c_ζ,

$$R \notin F_\sigma \iff (\exists n)\, R \restriction n \in V_\sigma.$$

Indeed, finitely many elements of $M_R = \bigcup_n c_{R\restriction n}$ all occur in the same $c_{R\restriction n}$ for some n. Therefore

$$F_\sigma = 2^\omega \setminus \bigcup_{\zeta \in V_\sigma} [\zeta]$$

is a Π^0_1 class. The definition applies to every R, even when M_R is not maximal, and does not assume totality of Φ^{J_R} on those parameters.

Constancy on $E_{M_G} \cap [\sigma]$ implies $G \in F_\sigma$. Otherwise, the two finite prefixes with incompatible outputs could be extended to two complete enumerations of M_G belonging to the same cone.

A 1-generic real belonging to a Π^0_1 class belongs to its interior. Applying genericity to V_σ, find $\xi \prec G$ with

$$[\xi] \subseteq F_\sigma.$$

Extend ξ further along G until $\operatorname{ran}(\sigma) \subseteq c_\xi$, and put $c = c_\xi$. This is possible because $M_G = \bigcup_n c_{G\restriction n}$ and $\operatorname{ran}(\sigma)$ is finite.

We claim that, for every maximal $N \supseteq c$, Φ is constant on $E_N \cap [\sigma]$. If not, two enumerations in that cone would have distinct outputs. There would then be finite prefixes $\tau_0, \tau_1 \succeq \sigma$, with values in N, such that $\operatorname{Split}_\Phi(\tau_0, \tau_1)$.

The set

$$D = c \cup \operatorname{ran}(\tau_0) \cup \operatorname{ran}(\tau_1)$$

is centered, as a finite subset of the same centered set N. By local universality there exists $\eta \succeq \xi$ with $c_\eta = D$. For every $R \succeq \eta$, the same prefixes witness $R \notin F_\sigma$, contradicting

$$[\eta] \subseteq [\xi] \subseteq F_\sigma.$$

This proves the claim.

Lemma 3, applied to c, σ, now provides a computable output g of Φ on a maximal solution. By hypothesis, g should be 1-generic.

No computable real is 1-generic: if g is computable, the computable set

$$U = \{(g \upharpoonright n)^\frown(1 - g(n)) : n \in \mathbb{N}\}$$

contains no prefix of g, but contains an extension of every prefix of g. This is the required contradiction. □

7. Consequence for Weihrauch reducibility

The preceding theorem already fixes one computable instance and rules out all uniform postprocessors. This immediately excludes an ordinary Weihrauch reduction by specializing its input to a computable oracle.

Regard 1-GEN as the problem which, given an oracle X, asks for a real that is 1-generic relative to X. FIP associates with every nontrivial family a maximal subfamily having the finite intersection property, represented by a sequence in which repetitions are permitted.

Corollary 6.

$$\text{1-GEN} \not\leq_{\mathrm{W}} \text{FIP}.$$

Proof. Suppose a pair of computable functionals Δ, Ψ witnesses 1-GEN $\leq_{\mathrm{W}}$ FIP. Fix the computable oracle $X = \varnothing$ as the initial instance. The preprocessor produces a nontrivial computable family

$$\mathcal{A} = \Delta(\varnothing).$$

For every maximal centered set of indices M and every $J \in E_M$, the sequence

$$(A_{J(n)})_{n \in \mathbb{N}}$$

is an FIP solution computable from J. Composing this transformation with the postprocessor Ψ and the computable parameter $X = \varnothing$, we obtain a functional Φ that produces an absolute 1-generic from all these enumerations. This contradicts Theorem 1. □

8. Representations and nonuniformity

The separation is for *ordinary* Weihrauch reducibility, not merely its strong variant: fixing the computable initial instance in the preceding proof absorbs the original input accessible to the postprocessor.

Indices and sets need not be in one-to-one correspondence: repetitions among the A_i are permitted. Maximality of a centered set of indices gives maximality of the corresponding family of sets, since any additional set that could be adjoined would have an index that could also be adjoined. This remains true when FIP is formulated using collections of at least two distinct members: the selected sets are nonempty, so such a maximal centered family is a valid FIP solution. The converse is not needed. We use only the computable conversion

$$J \longmapsto (A_{J(n)})_{n \in \omega},$$

not its converse. In particular, no decision procedure for equality of computable sets is required; cf. the discussion of solution presentations in [4, Section 1.1].

The splitting argument uses the union of finite input ranges only when those ranges are contained in a *common* centered set. The finite constants chosen in the proof, including σ and c_ξ, need not be found uniformly from $\mathcal{A}$; once chosen, they can be hard-coded in the

computation of the final real. Thus the argument does not conflict with the nonuniform computability-theoretic equivalence proved in [2]. Their earlier observation concerning the failure of uniformity is an important antecedent of the present separation.

Dzhafarov and Mummert also asked in [5, Question 9.10.26] whether $\mathsf{RCA}_0 \vdash \mathrm{2IP} \to \text{1-GEN}$. This question is answered affirmatively in a companion manuscript [3]. There is no conflict between this reverse-mathematical result and the Weihrauch nonreducibility established here: provability of an implication over RCA_0 does not, by itself, yield a Weihrauch reduction. In particular, the companion proof uses a nonuniform distinction between an already generic path and a case in which the jump of the original instance can be decoded.

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Mathematics Division, School of Science and Technology, University of Camerino, 62032 Camerino, Italy

Email address: patrizio.cintioli@unicam.it